

A

Project Report on

SHETISAKHA

An IoT based Smart Automation System for Agricultural Activities

Submitted in fulfilment of the requirements

for the award of

Diploma in

Information Technology

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ShetiSakha: An IoT based Smart Automation System for Agricultural Activities

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We hereby certify that the work which is being presented in the project is entitled. **“ShetiSakha: Iot based Smart Automation System for Agriculture”** by us in partial fulfillment of the requirements for the award of Diploma in Information Technology, Government Polytechnic Nagpur submitted to Department of Information Technology is record of our work carried out during the Academic session 2023-24 guided by Prof. L.D.Vilhekar and Prof. P.Saktel.

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Project Guide

Acknowledgement

Today on completion of this project, the people we need to thank the most who have helped us throughout the making of the project dissertation and without whose support the project would not have seen the light of the day.

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Abstract

India is a global agricultural powerhouse. In India, agriculture is the primary source of income for the majority of the population. Currently, around 70% of people live in villages whose main source of food is the revenue generated from farming.

According to a report published by the World Bank, the challenges to India's overall development and the improved welfare of its rural poor were namely, raising agricultural productivity per unit of land, ensuring that agricultural growth responds to food security needs, and reducing rural poverty.

India's agricultural sector faces challenges primarily due to the prevalence of repetitive manual labour and time-consuming tasks. The country's vast agricultural landscape relies heavily on traditional farming methods, where much of the work, from sowing to harvesting, is done manually. The lack of mechanization means that tasks that could be efficiently completed with automation take a significant amount of time and effort. This reliance on manual labour not only leads to inefficiencies but also limits the scalability of agricultural operations.

Additionally, the dependency on labor-intensive methods often results in delayed or untimely completion of tasks, impacting crop cycles and overall productivity. The need for modernization and automation in these repetitive tasks is crucial to alleviate the burden on farmers, enhance efficiency, and boost agricultural output, ensuring India's food security and economic growth in the long run.

By adopting advanced technologies like precision farming, automated machinery, and smart sensors, agricultural practices can become more efficient and yield higher outputs. Automation minimizes human error, optimizes resource utilization, and enables timely and precise operations, leading to increased crop yields. Moreover, it addresses labor shortages, allowing farmers to focus on strategic decision-making and innovation. In addition to bolstering agricultural production, automation also promotes eco-friendly practices, conserves resources, and ultimately contributes to the country's food security and economic development.

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Chapter 1

Introduction

ShetiSakha is the initiative of the students of Government Polytechnic Nagpur to help the farmers to plough, seed and irrigate the field without manual load. By using the system the farmers can remotely control the machine via an app. The model will automate the lanes and traverse through the farm without any hustle.

The idea of automated agricultural machines originates from the need to enhance efficiency, reduce labour, and meet the growing demands of modern farming practices.

1.0.1 Existing System

Currently the existing system for taking care of the Current agricultural practices in India face challenges related to ploughing, sprinkling, and seeding methods. Manual ploughing, often utilizing oxen, can be labor-intensive and time-consuming, impacting productivity. Traditional sprinkling techniques, relying on manual methods, result in uneven water distribution, affecting crop growth. Additionally, seeding processes lack mechanization and precision, leading to inconsistent results. These challenges highlight the need for modernization and technological advancements in India's agricultural sector to improve efficiency and overall productivity.

1.0.2 Motivation

This project is inspired by a survey conducted with the farmers regarding the work they do and the difficulties that are faced in traversing the whole farm multiple times. The farmers gave a brief description of the issues faced by them and after seeing live farming and the treacherous work the team decided to integrate a sensor using an Arduino UNO for the given problem statement.

1.0.3 Proposed System

The proposed system is a prototype model that deals with the challenges faced by the farmers while traversing the farm to plough, to sow seeds and it irrigate. It aims to provide a means for hassle-free agricultural work.

1.0.4 Features of Proposed System

- The system provides a weather report of any particular city, displaying the temperature in Celsius, along with winds, pressure and humidity readings
- The system can be remotely operated via an app by sending values to turn the machine on or off.
- The system sends signals to the machine for putting the plough down, seeding or turning the pump of the sprinkling on.
- The system can also send path as an input to the machine through the android application.
- It can be placed on auto mode once the path is memorized and is to be repeated from starting to the endpoint.
- It detects obstacles in front of the machine, that can result in crash and halt the machine accordingly.

1.1 Software Engineers Approach

Software lifecycle is considered as a subset of system development life cycle. Out of many models for such processes every describing approach to a spread of tasks or activities that happen throughout the method. Each process model follows a particular well defined life cycle in order to ensure success in the process of software development.

A software process model is an abstract representation of a process. Waterfall approach is a classical and oldest process model to be introduced and adopted widely in software system engineering to confirm success of the project. This model mainly emphasizes exhaustive planning in the early stages thus it helps to reduce design flaws before they develop. In the waterfall approach, the complete method of code development is split into no. of distinct phases. Phases in waterfall model are:

1. Requirement Specification Phase

2. System and Software Design
3. Implementation and Unit Testing
4. Integration and System Testing
5. Operation and Maintenance

The waterfall model is a logically sequential design process. Here each phase has a distinct role and well-defined set of activities and hence the waterfall model is characterized as one should move to a next phase only when its preceding phase is completed and perfected. All the methods and processes undertaken in the waterfall model are distinct as well as clearly visible.

1.2 Stages in Waterfall Model

1.2.1 Requirement Specification and Definition

All possible components required for building the system to be captured in this phase including the hardware requirements, interface requirements, special power requirements, software tools, and other necessary components. The proposed system's major requirements are the Arduino UNO board, sensor, Bluetooth HC05, Arduino software for its programming and Android Studio.

1.2.2 System and Software Design

It includes low-level design and high-level design. Low-level design defines the major components like System UI. In contrast, the high-level design examines the software components and their specification defined in the low-level design stage have detailed information about how each component is implemented.

1.2.3 Implementation and Unit Testing

On receiving system clear and concise design documents the system is first developed in various small subprograms called units/ modules which are integrated further in the next phase.

1.2.4 Integration and System Testing

Software testing is an integral and crucial phase of the software development process. Purpose of the testing is to determine whether the software meets the specified requirements or not. Early testing will resolve a need for redesign, thus ultimately reducing cost. Initially each module is tested individually. Finally, all modules are integrated and tested collectively to ensure that the system meets functional requirements.

1.3 Organisation of Dissertation Report

The rest of dissertation report is organised as follows:

Chapter 2 shows software requirements and related proposed work. Chapter 3 specifies complete system design necessarily having UML's and architecture. Chapter 4 introduces technical specifications required for the set up of implementation of the proposed system. Chapter 5 is devoted to cost and effort estimation required for completion work. Chapter 6 briefly deals with complete implementation details involved with help of certain assumptions. Chapter 7 specifies testing of different modules and systems collectively. Complete result analysis and performance evaluation as well as comparison of proposed systems is discussed in Chapter 8 and finally Chapter 9 concludes the proposed work and short discussion on future enhancements related with the topic.

Chapter 2

Software Requirements Specification

2.1 Introduction

ShetiSakha is the initiative of the students of Government Polytechnic Nagpur to help the farmers perform multiple operations like ploughing, sowing and sprinkling without any hassle. By using the system, the farmers can remotely control the machine and traverse through the field via an app. The model will automate the lane navigation by two ways- path memorizing and draw-line mode.

The idea of automated agricultural machines originates from the need to enhance efficiency, reduce labour, and meet the growing demands of modern farming practices

2.1.1 Purpose

The purpose of the project is:

- To help the farmers provide multiple operations like ploughing, seeding, and irrigating.
- To help farmers reduce the manual work by automating field traverse.
- To able farmers to operate the mobile via the app remotely
- To increase the yield per piece of land

2.1.2 Project Scope

The system will have a hardware component machine is to be placed on the farm, and then using the mobile app it will perform the three operations of ploughing, seeding, and sprinkling. It will traverse through the farm form point A to B by manual controls and by memorizing this path it can be repeated for point A to B.

2.1.3 Design and Implementation Constraints Design

For the development of the proposed system following is the requirement:

- Arduino UNO
- Arduino IDE
- Android Studio
- Ultrasonic Sensor
- HC05 Bluetooth Module
- Relay Module

2.2 System Features

- Save time and resources
- Efficiency
- Reduce manual work
- More convenient

2.3 Hardware Interfaces

- Arduino uno
- relay module
- Bluetooth module HC 05
- Pump
- ultrasonic sensor
- motor driver
- Arduino uno sensor shield 3
- DC motor
- Fan

2.3.1 Software Interfaces

- Android studio
- Arduino IDE
- SQLite Database

2.4 Software Quality Requirements

Following table shows various quality attributes and it level in order to test quality of the system:

Attribute	Performance
Availability	Controls directions of the machine remotely. Also controls ploughing, seeding and sprinkling mechanisms
Maintainability	The system has path memorizing and draw-line mode to reduce human interaction
Usability	Easy to use
Aesthetic	The system is delivered in a compact and easy to install box

Table 2.4: Software Quality Attributes

Chapter 3

System Design

3.1 UML Diagrams

UML stands for Unified Modelling Language. It represents a unification of the concepts and notations. UML creates a common language for creating models of object-oriented computer software. It is a standard way for system representation and interaction. It describes the objects, attributes, and relationships necessary to represent the concepts of UML within a software application.

3.1.1 Use Case Diagram

Use-case diagrams are used to define functions of the system and to capture and analyse the functional requirements of an application system. A use case model is a collection of use cases which specify the behaviour of a system.

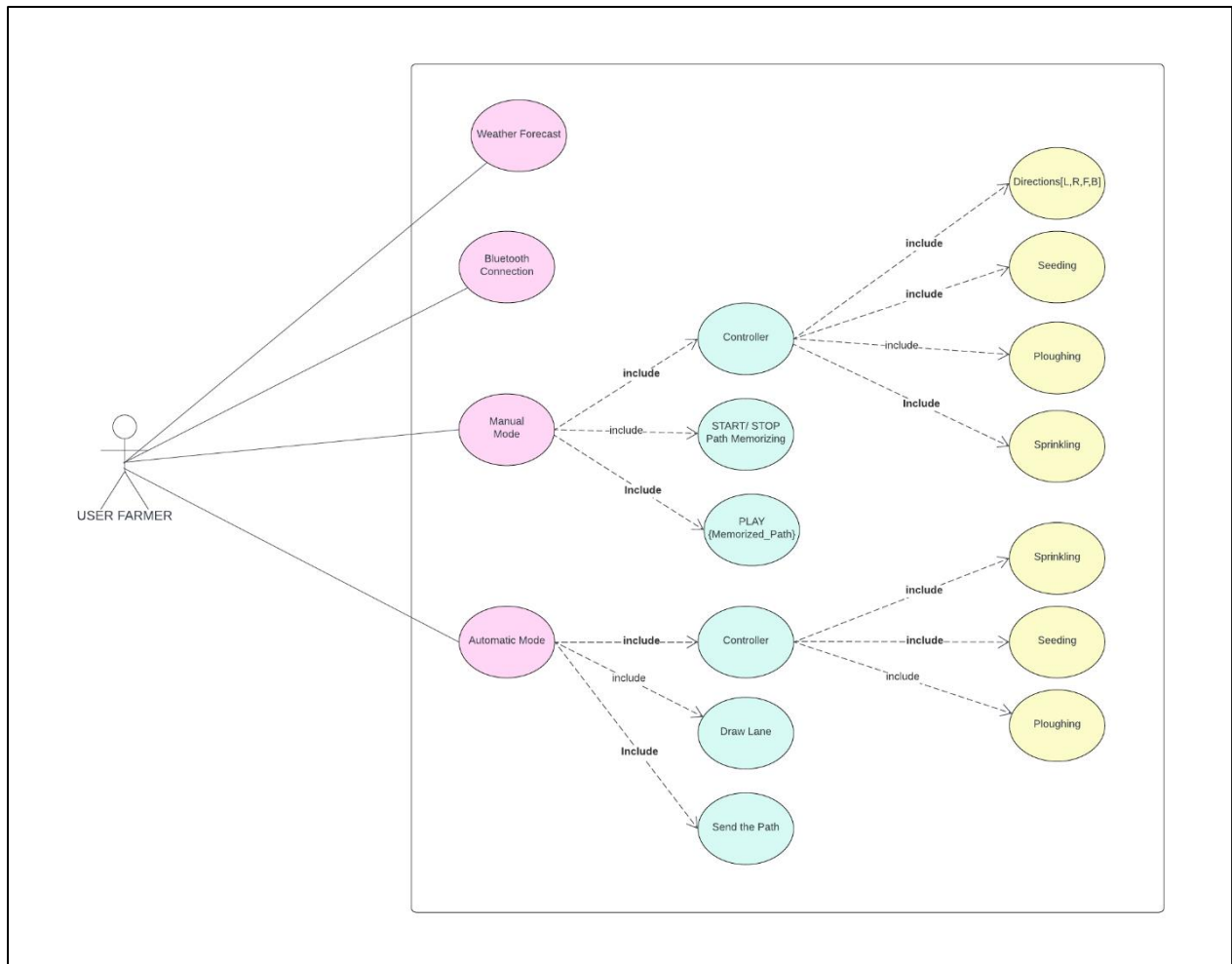


Figure 3.1.1: Use-case Diagram

3.1.2 Activity Diagram

An activity diagram shows the flow of activity to activity. Activity diagram is just a special kind of diagram and shares the same common properties as do all other diagrams. Activity diagrams commonly contain activity state and action states, transition and objects. Activity diagram is used to model the behaviours of a system, and the way in which these behaviours are related in an overall view of the system.

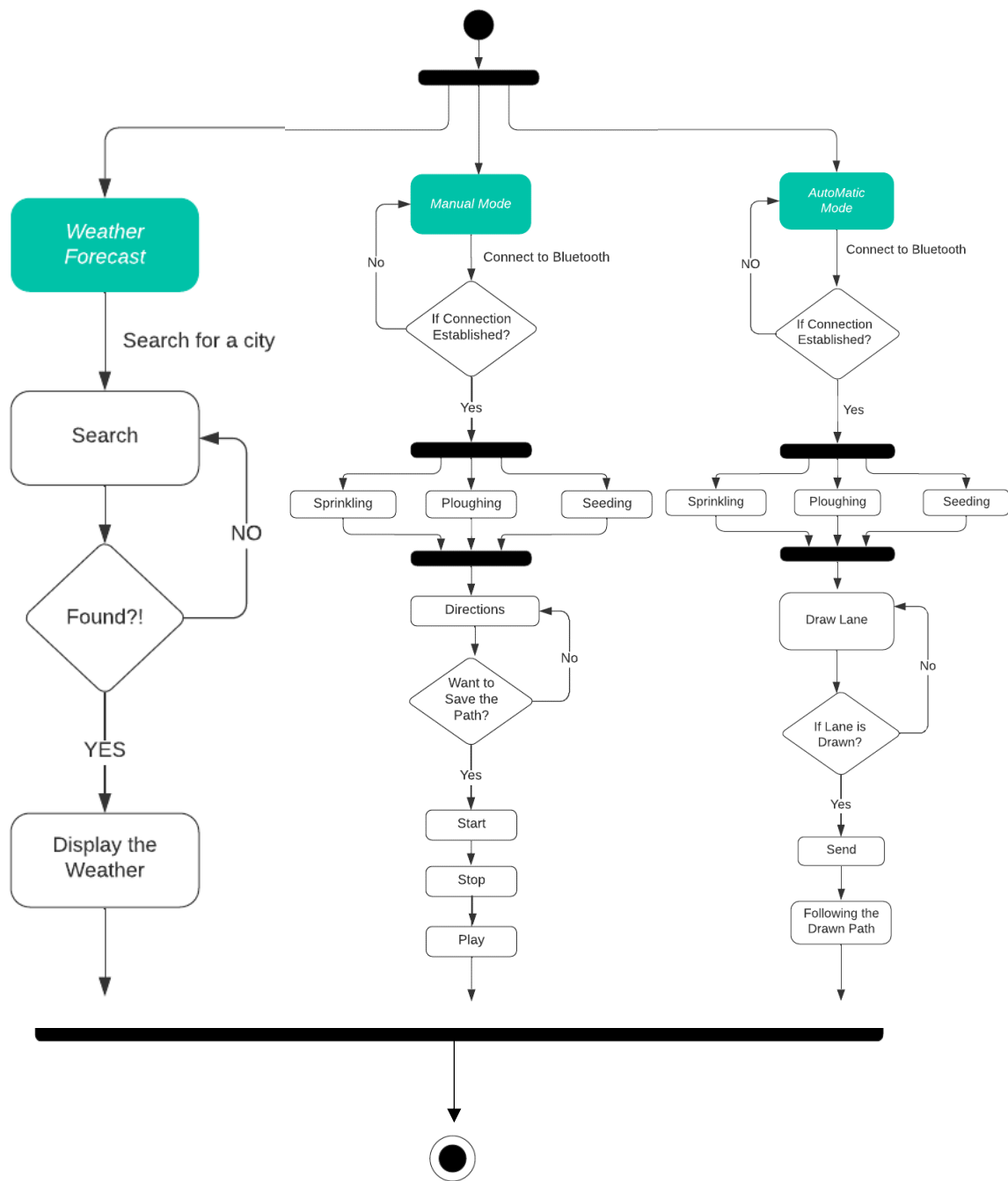


Figure 3.1.2: Activity Diagram

3.1.3 State Diagram

A state diagram shows a state machine, emphasizing the flow of control from state to state. State is a condition or situation in the life of an object during which it satisfies some condition, performs some activity or waits for some event. Event is the specification of a significant occurrence that has location in time and space. State diagram is just a special kind of diagram and shares the same common properties as do all other diagrams.

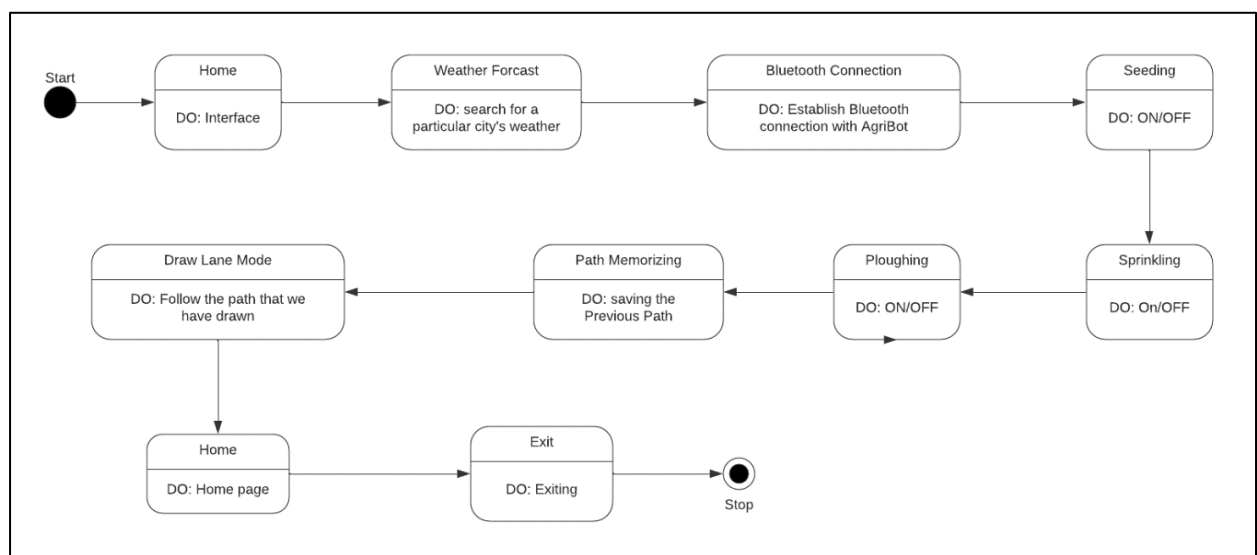


Figure 3.1.3: State Transmission Diagram

Chapter 4

Technical Specifications

4.1 Arduino Uno-Sensor Shield

1.Connectivity Simplification:

- The sensor shield simplifies the process of connecting sensors and other devices to an Arduino Uno by providing pre-defined connectors.

2.Convenient Interface:

- It provides a convenient interface for various sensors and components through labeled headers and connectors.

3.Reduced Wiring Complexity:

- Reduces the complexity of wiring and makes the setup more organized by allowing sensors to be plugged into corresponding slots.

4.Ease of Prototyping:

- Facilitates rapid prototyping by eliminating the need for extensive wiring, allowing quick experimentation with different sensors and components.

5.Incorporates Additional Features:

- Some sensor shields include extra features like buttons, LEDs, and more, providing additional input/output options in projects.

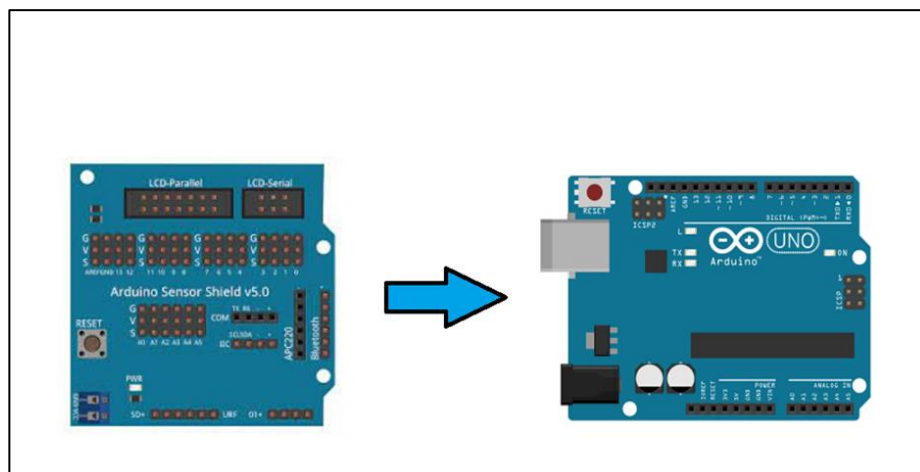


Figure 4.1. Arduino Uno connection with Arduino Sensor Shield

4.2 HC-05 Bluetooth Module

The Bluetooth module of ShetiSakha is a crucial part of the system. It enables data transmission and control commands over Bluetooth, making it a common choice for projects involving wireless control, Bluetooth-based automation.

The HC-05 Bluetooth module is a highly adaptable component widely utilized in automation projects and applications. Its primary function is to establish wireless connections between devices, enabling data exchange and control commands over Bluetooth technology.

To control the module , we have used an Arduino board. To connect an HC-05 Bluetooth module to an Arduino, follow these steps: First, power the HC-05 by connecting its VCC and GND pins to the Arduino's 5V and GND pins. Establish serial communication by connecting the TX pin of the HC-05 to the Arduino's RX pin and vice versa. Finally, use the SoftwareSerial library to communicate with the HC-05 in your Arduino code. This allows you to send and receive data over Bluetooth, making your Arduino accessible via Bluetooth from a paired device.

The Bluetooth control module of the ShetiSakha system plays vital role as it ensures the wireless communication through the commands given by the software and transmit signals to the microcontroller making it convenient and easy to operate.

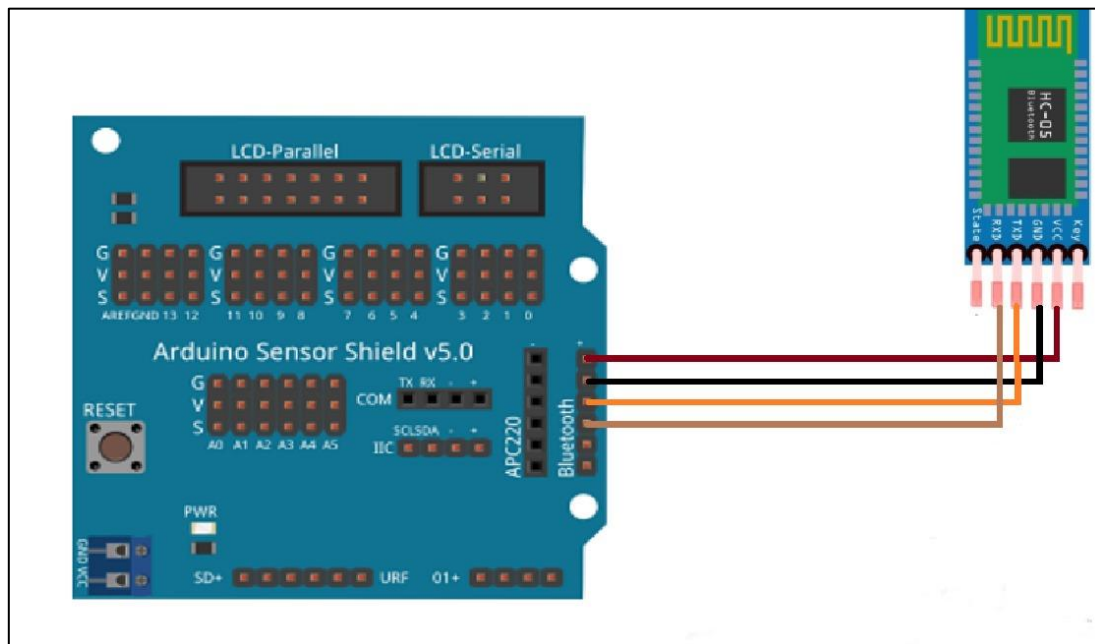


Figure 4.2 HC-05 Bluetooth control module circuit

4.3 HC-SR04 Ultrasonic Sensor Module

The ultrasonic sensor module in the ShetiSakha system detects the obstacle in the field. It sends out sound waves and measures how long it takes for the sound to bounce back. By calculating this time, it determines the distance to an object in front of it, making it useful for applications like measuring distances or detecting obstacles. The HC-SR04 ultrasonic sensor has four pins. VCC is the power supply pin (usually 5V), Trig is the trigger to initiate distance measurement, Echo receives the echo signal, and GND is the ground connection. By sending a trigger signal and measuring the time it takes for the echo signal to return, the sensor calculates distances, making it useful for a variety of applications like obstacle detection and distance measurement.

To connect an HC-SR04 ultrasonic sensor with an Arduino, connect the VCC pin of the sensor to the 5V output on the Arduino, the GND pin to the Arduino's ground, the Trig pin to a digital output pin (e.g., D2), and the Echo pin to another digital input pin (e.g., D3). Ensure you have a common ground between the sensor and the Arduino for reference. Then, in your Arduino code, use the specified trigger and echo pins to send a trigger pulse, measure the echo duration, and calculate distances based on the time it takes for the sound wave to bounce back. This allows you to create applications like distance measurement and obstacle avoidance with the HC-SR04 and Arduino.

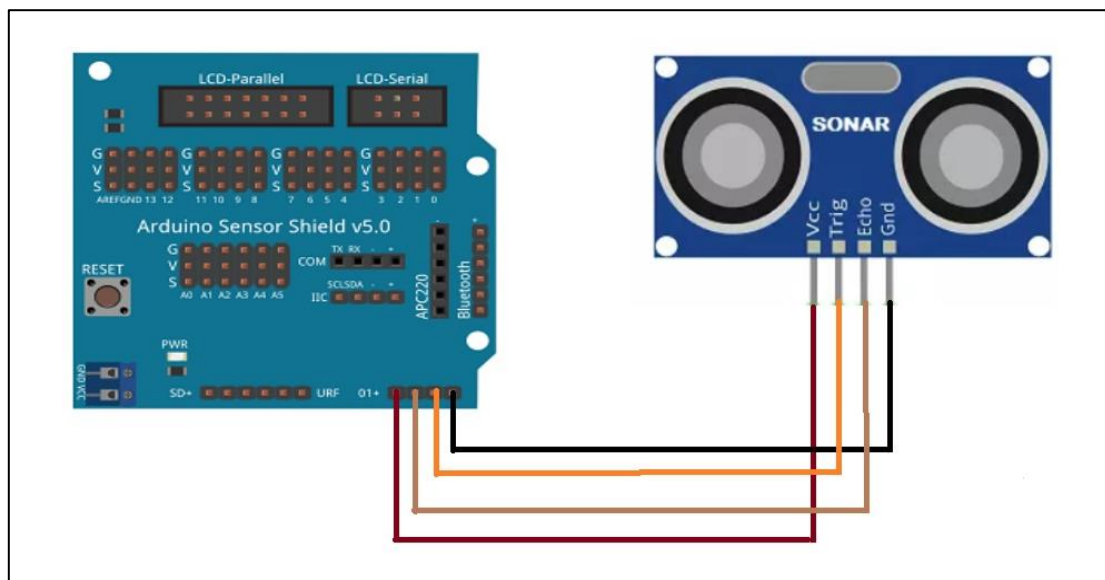


Figure 4.3: Ultrasonic sensor module (HC-SR04)

4.4 Relay Module

The relay module enables to control devices like motors, or appliances by allowing or interrupting the flow of electricity, making it a valuable component for automation. In ShetiSakha, with the help of relay module all the appliances are controlled by the high and low command given by the microcontroller. A relay module typically has four important pins: VCC for power, GND for ground, IN for the control signal, and COM, NO, and NC for switching connections. It allows a low-power signal from a microcontroller to control high-power devices by connecting or disconnecting the COM pin to NO or NC, depending on the control signal, making it a key component in applications like turning on/off motors and other appliances.

To connect a relay module with an Arduino, connect VCC and GND for power, and wire the control pin (usually "IN") to a digital output pin on the Arduino. This allows you to use the Arduino to switch high-power devices on and off using the relay, commonly applied in automation and robotics.

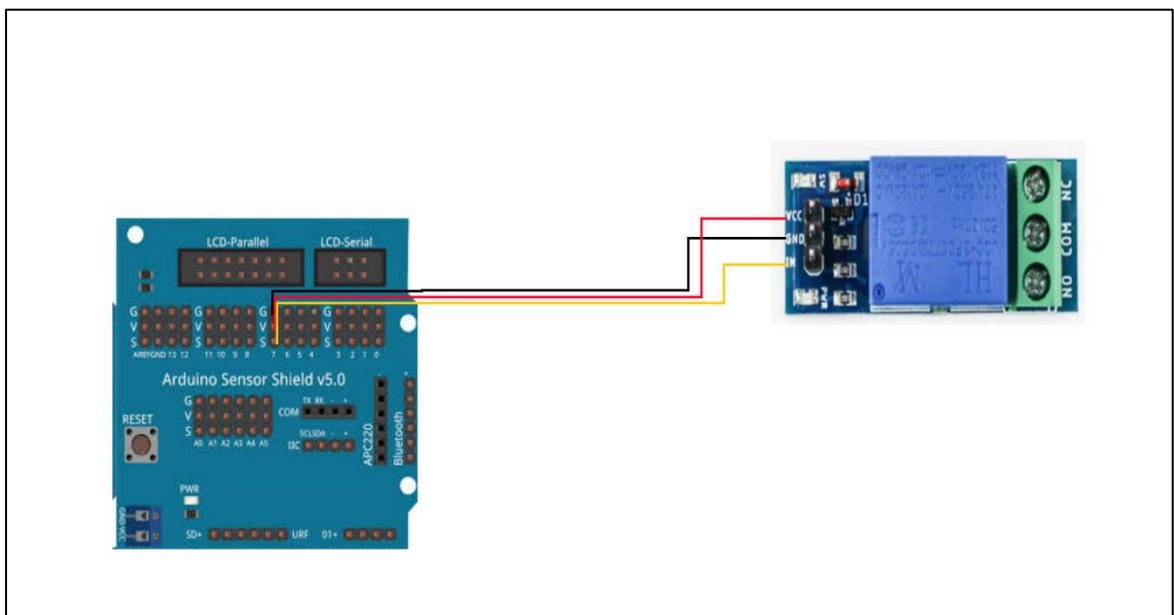


Figure 4.4. Relay Module

4.5 L298 Motor Driver

The L298 driver in the ShetiSakha system drives the robot wheels. It allows to control the direction and speed of DC motors. It takes input signals to control the motor rotation and provides the necessary power for motor operation. It's commonly used in robotics and specially in motor control. The L298 motor driver is like a remote control for electric motors. The commands are send by the microcontroller establishing connection with Bluetooth module and the motors perform the function of forward, backward, right, left or stop. It needs its own batteries to work, and you connect the motors to it. It's handy for building things like robots or small vehicles that need motor control.

In L298 Driver ,connect the input pins (IN1, IN2, IN3, IN4) from the motor driver to digital pins on the Arduino. Next, attach the enable pins (ENA and ENB) from the motor driver to other digital pins on the Arduino. The motor driver needs its own power supply, so connect its motor power supply (Vs and GND) to a separate battery or power source. Don't forget to connect the motors to the motor driver's outputs (OUT1, OUT2, OUT3, OUT4). To make everything work smoothly, connect the ground (GND) of your Arduino to the motor driver's ground (GND). This setup lets you control your motors using the Arduino by establishing serial communication with Bluetooth module.

Arduino can work with an L298 motor driver to control DC motors. Here's a simplified overview of how it works:

- **Connection:** Connect the Arduino to the L298 motor driver using digital output pins for motor direction (IN1 and IN2) and a PWM (Pulse Width Modulation) pin for motor speed control (EN).
- **Programming:** Write an Arduino program (sketch) that sends signals to the IN1 and IN2 pins to set the motor direction (forward, backward, or stop). You can also use the `analogWrite` function to control the EN pin and adjust the motor speed by varying the PWM duty cycle.
- **Power Supply:** Connect a suitable power supply to the VCC and GND pins of the L298 to provide power to both the motor driver and the motors.
- **Feedback:** Optionally, you can use sensors, such as encoders or limit switches, to provide feedback to the Arduino, enabling it to make decisions based on the motor's position or status.

- **Control:** The Arduino processes your code and sends control signals to the L298 motor driver, which, in turn, drives the connected motors accordingly.

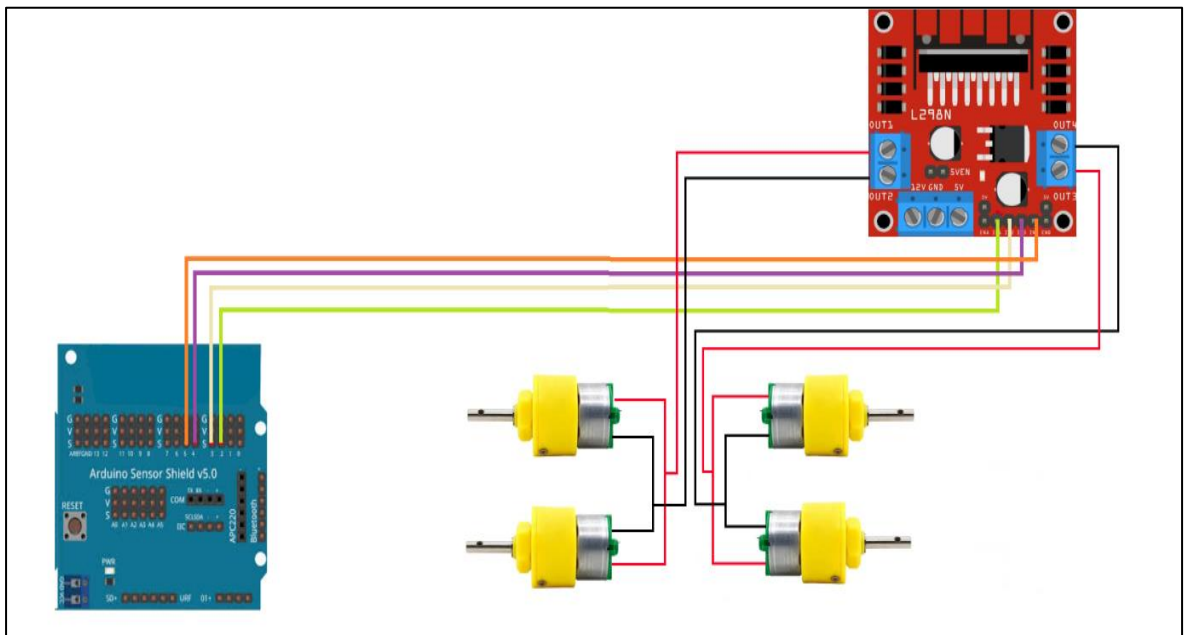


Figure.4.5 Connection of DC gear motors with L298 Motor Driver

4.6 Ploughing

In ShetiSakha gearbox lifter is used , in agriculture it is a component that helps control the depth and speed of ploughing. It's like the transmission in a car, adjusting the power to the plough blades. This gearbox ensures efficient and precise ploughing when used with ShetiSakha Below is a a simplified overview of how your hardware mechanism for ploughing, with a gear box and controller using relays and Arduino,

- **Arduino Control:** The Arduino serves as the central controller for your ploughing system.
- **User Input:** You can manually input the desired depth and speed settings into the Arduino.
- **Relay Control:** The Arduino sends control signals to relays, which adjust the gear box settings according to your input.

- Gear Box Adjustment: The relay switches modify the speed and torque of the ploughing equipment based on your manual settings.
- Ploughing: With the gear box adjusted, the ploughing equipment operates at the specified parameters.

This setup allows for manual control of the ploughing process, with the Arduino and relays helping you fine-tune the gear box settings for efficient and precise ploughing.

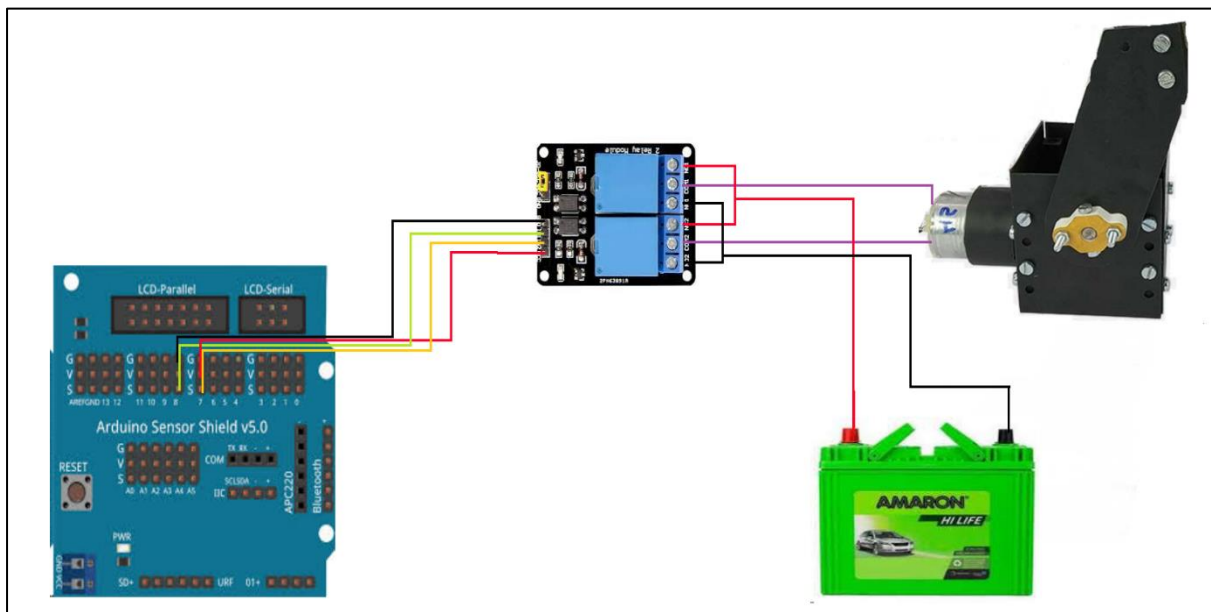


Figure 4.6. Ploughing operation

4.7 Sprinkling

In ShetiSakha brass sprinklers are used for the sprinkling .Brass sprinklers are durable and corrosion-resistant due to their construction material. They provide an effective and reliable method for irrigating crops ,farm , offering adjustable watering patterns to suit various landscaping needs.

A 12-volt submersible motor is like a special electric motor that can work underwater. It's often used in brass nozzle to give pressurized spray. This motor is sealed to keep water out on a 12-volt power source, making it safe and useful for a variety of tasks in wet conditions.

Certainly! In simple terms, for water supply in a ShetiSakha (agricultural field), we've used a 12-volt motor, a relay module, and an Arduino:

- **12-Volt Motor:** This motor runs on a 12-volt power source and is responsible for pumping water from a source to where it's needed.
- **Relay Module:** The relay acts like a switch that your Arduino controls. It turns the motor on and off, allowing you to start and stop the water pump.
- **Arduino:** Your Arduino is like a mini-computer that's programmed to decide when to activate the relay, which, in turn, starts or stops the motor to control the water supply.

By combining these components, you've created an automated system that uses the Arduino to manage your water supply in the ShetiSakha, making more efficient and reliable.

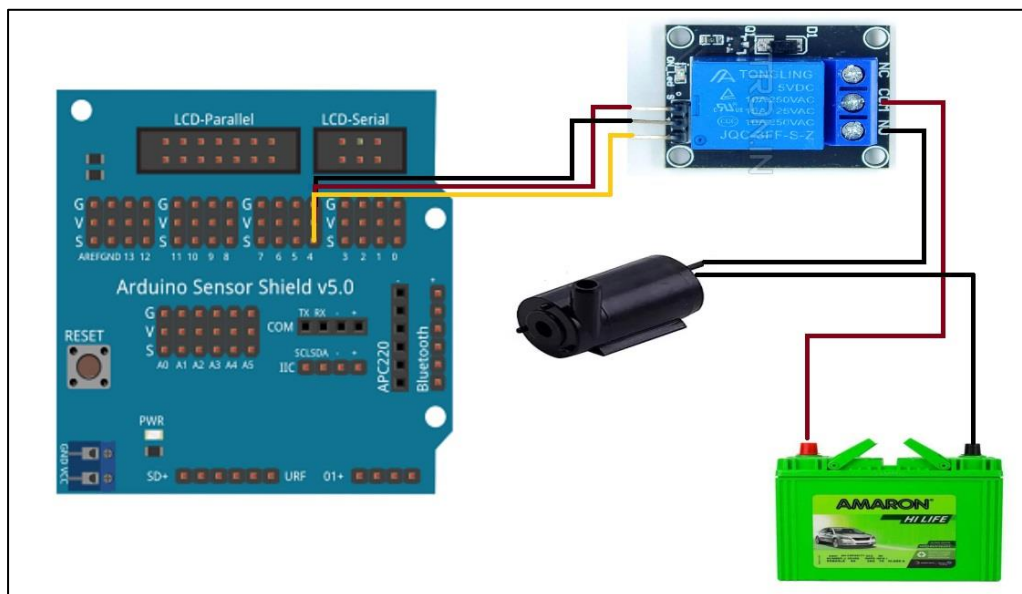


Figure 4.7 Sprinkling operation

4.8 Seeding Mechanism

- **Components:** Gather a 5V DC motor, a relay module, an Arduino, a power supply, and your seeding mechanism.
- **Motor and Power:** Connect the DC motor to a 5V power supply.

- **Relay Module:** Connect the relay module to your Arduino to control the motor's on/off using digital pins.
- **Programming:** Write an Arduino sketch to control the motor via the relay.
- **Assembly:** Attach the motor to your seeding mechanism.
- **Power Up:** Connect power to the motor and relay module and upload your Arduino sketch.
- **Testing:** Test and calibrate the seeding mechanism.
- **Mount:** Secure the mechanism to your ShetiSakha or equipment.

This setup allows your Arduino to automate seed dispensing

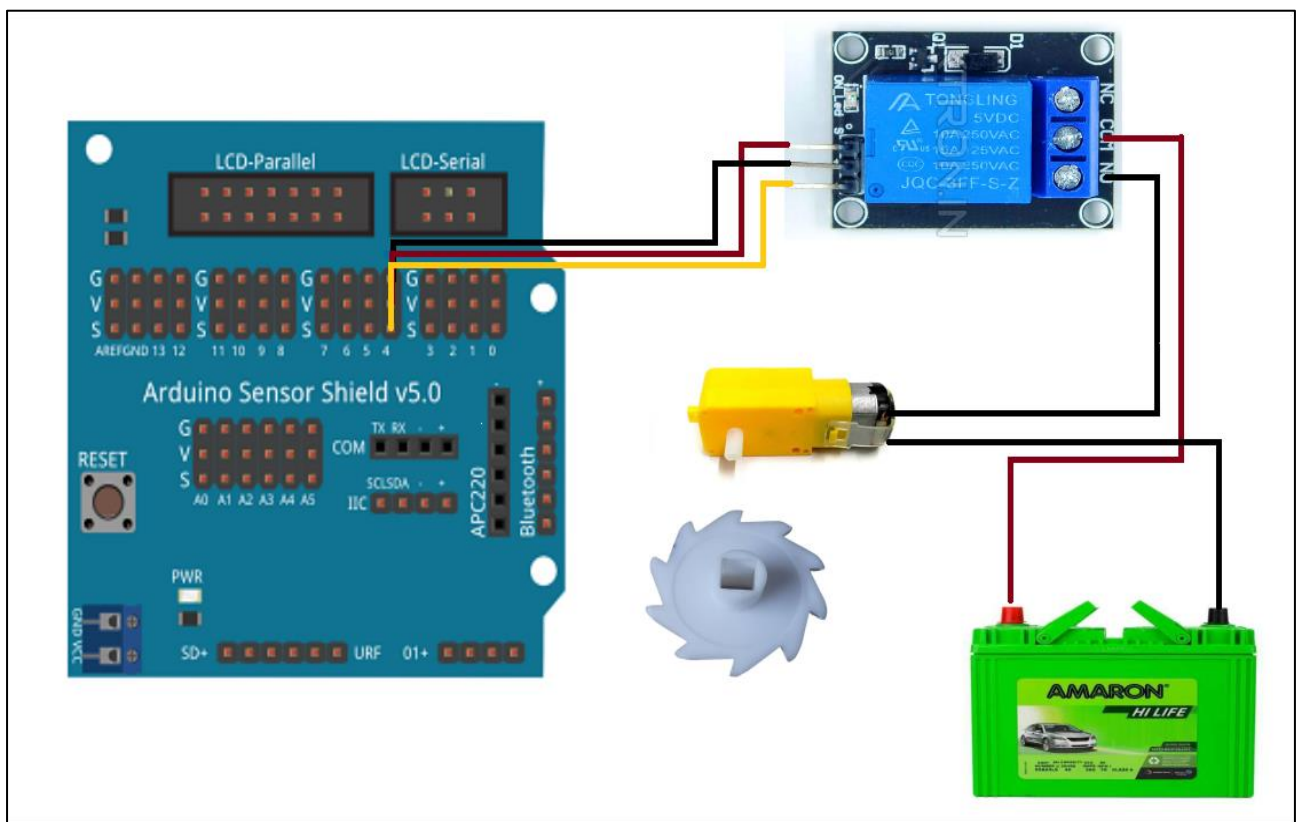


Figure 4.8 Seeding Mechanism

Chapter 5

Project Estimate and Schedule

5.1 System Estimation plan

Sr. no.	Name	Description	Timeline	Remark
1	Communication	Communication with Client to know the requirement of the app	18/7/2023 to 24/7/2023	Gathering Requirement through Communication with the client
2	Requirement	Complete Specification of system	25/7/2023 to 31/8/2023	A Detailed document should be there for each requirement
3	Information Gathering	GUI design, Program Specification	01/08/2023 to 12/08/2023	Make the circuit diagrams for modules
4	Planning and Modeling	Making plan to build the system	13/08/2023 to 17/08/2023	Decide on different modules and how they are interacting
5	Design	Design the module created	18/08/2023 to 31/08/23	GUI design should be completed also different screens needed should be decided
6	Coding and implementation	Code of system, Implementation of different modules	01/09/2023 to 01/10/2023	Write code for different modules
7	Testing	Test the different module together	02/10/2023 to 15/10/2023	System is to tested using different test strategies

5.2 Estimation

The creation of ShetiSakha demanded thorough planning through a comprehensive estimation procedure to gauge the project's scope, labour, and timeline. Our initial step involved assessing the product's size by describing its components and specifications. Subsequently, we calculated the necessary effort in man-months, factoring both hardware and software elements. Lastly, we determined the project schedule in calendar months, accounting for variables like resource availability, testing, debugging, and potential setbacks. This systematic estimation process played a pivotal role in guaranteeing the project's timely completion.

5.2.1 LOC based Estimation Effort

Factors for the estimation of effort for development of ShetiSakha include size of the project, the complexity of the software and hardware components, the expertise of the team, and the time required for testing and debugging. One approach to estimating the effort required is using the Lines of Code (LOC) method. This involves calculating the total number of lines of code required for the project and then using industry-standard metrics to estimate the effort required. The LOC method is often used in software development to estimate the time and resources required for the project.

The assessment of the size, effort, and schedule for the development of ShetiSakha can follow a systematic procedure. To estimate the size of the project, we must consider the multitude of features and functionalities it encompasses. Elements such as the diversity of tasks it can perform, the intricacy of its control interface, and its compatibility with other modules all contribute to the project's size.

Once the effort required has been estimated, the schedule can be determined based on the available resources and the project deadline. A calendar month estimation can be made based on the effort required, as well as the number of team members available to work on the project. In summary, the estimation of the size, effort, and schedule of the ShetiSakha terminal control module can be done using the LOC method, which takes into account various factors such as

project size, complexity, and team expertise. This estimation can help ensure that the project is completed on time and within budget.

Sr. no.	Function	Line of code
1	Weather	370
2	Manual mode	960
3	Automatic mode	1732
4	Navigation Drawer	230

1. Estimation Effort [PM]

$$PM = a[KLOC]^b$$

$$PM = 3.0[3.292]^{[1.12]PM}$$

$$= 12.94 \text{ person-month}$$

2. Development time[T day]

$$\text{Estimated Time} = \text{Estimation Effort} / \text{No of person}$$

$$= 12.94 / 5$$

$$= 2.58 \text{ months}$$

5.3 Schedule

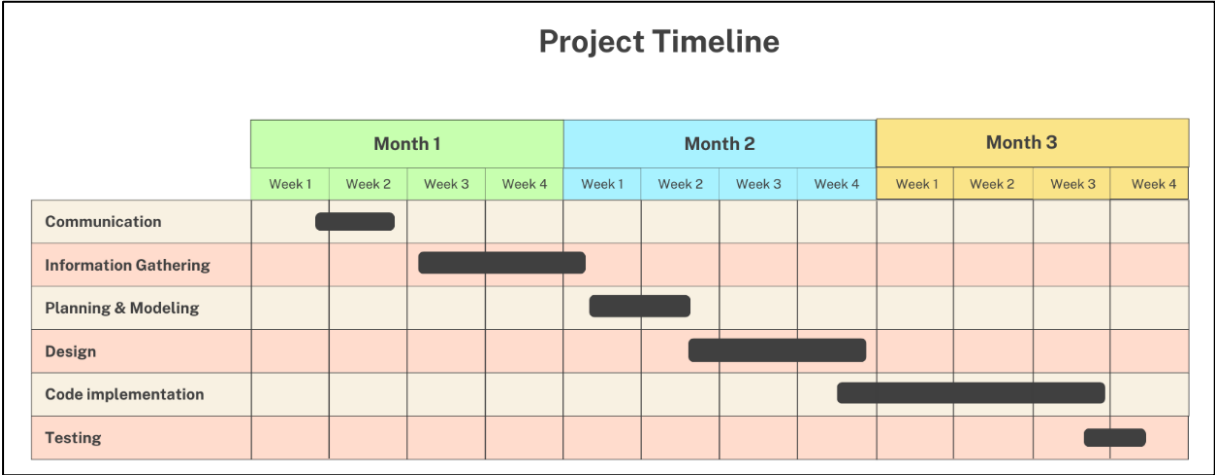


Fig 5.3: Gantt chart

Chapter 6

Software Testing

6.1 Introduction

Software testing is the process of evaluating and verifying that a software product or application does what it is supposed to do. Software Testing is the method of execution of a system with the decision of finding errors. Software testing is an investigation aimed at evaluating a certain attribute and capability of a program or system and predict that it meets its required specifications or not. Testing is one of the means to detect software failures so that defects can be discovered and corrected as early as possible. Testing ensures that all functions in software are properly working very well under all conditions and it establishes that performance doesn't perform properly under specific conditions. Software system to widen it typically includes exhaustive checking of code along with execution of that code under various environments and conditions as well as examining the various aspects of code. In today's culture of software development, a testing process is organised separately and parallel with the development process.

Software testing affects quality of software packages directly as a result of software may be used to correct the process by which software is developed. Software testing is a trade off between budget, time and quality. In simple software Testing is defined as questioning a certain product in order to evaluate it.

6.2 Test Cases

6.2.1 Definition

A test case may be of conditions or variables under which a tester will determine whether a system under test satisfies requirement or works correctly .

6.2.2 Fields in Test Case

- **Test Case ID**
- **Unit to Test** : What to be verified
- **Prerequisite**
- **Test Data** : Variables and their values
- **Test Procedure**
- **Expected Results**
- **Actual results**
- **Status**
- **Remarks**

6.3 Unit Testing

Testing is performed by the developers before the system set-up is transferred to the testing team to execute the test cases by using stubs and drivers. Unit testing is done by the respective developers to whom a module is assigned. The developers use test data that is separate from the test data of the quality assurance team. The purpose of the unit testing is to separate each part of the program and ensure that individual parts are working correctly in terms of requirements and functionality. Performing unit testing on the double guard system following modules are tested individually.

○ Unit Testing Limitations

Testing cannot capture each and every error in an application . It is not possible to evaluate and test every execution path in every software application .

6.4 White Box Testing

In White box testing the detailed structure of internal logic , structure of the code is implemented. White Box testing is also called glass-box testing or open box testing. White Box testing is the application the tester needs to have good domain knowledge as well as the tester should be aware about the internal working of the code loops. The tester needs to have a look inside the source code and search out which unit of the code is behaving inappropriately.

6.5 Black Box Testing

Testing method while not having any data of internal working of the application is Black Box Testing. The tester is a completely new one to the system architecture. Black box test a tester will interact with the systems user interface providing inputs and examining outputs while not knowing however and where the inputs are worked upon.

6.6 Module Testing

6.6.1 Pre - processing

Test Id	Input	Expected O/P	Actual O/P	Remarks
1	Press sprinkling On / Off button	Turn On / Off Water pump to sprinkle as per values sent .	Turn On / Off Water pump to as per the values sent .	Pass
2	Press seeding On / Off button	Seeding operation activated or deactivated as per user input	Seeding operation activated or deactivated as per user input	Pass
3	Press ploughing On / Off button	Ploughing operation activated or deactivated as per user input	Ploughing operation activated or deactivated as per user input	Pass

4	Test responsiveness of remote control buttons	Buttons respond promptly to user input and the bot follows directions	Buttons respond promptly to user input and the bot follows directions	Pass
5	Start recording path by giving direction via remote, stop recording and play the path	The bot follows the path during playback	The bot follows the path during playback	Pass
6	Draw the path in automatic mode and send path	The path is received and followed by bot	The path is received and followed by bot	Pass
7	Introduce obstacles in the path of machine during operation	The bot detects obstacle and stops	The bot detects obstacle and stops	Pass

Chapter 7

Deployment and Maintenance

7.1 Installation

- Android phone
- Agribot

7.2 Execution

ShetiSakha's execution phase revolves around empowering users to effortlessly interact with the agribot. Once the system is deployed and tested, it is ready for execution. The execution process involves the user opening the application. Once they have opened the application, they will be able to access the various functionalities of the system such as checking weather, controlling bot via manual mode or automatic mode and adjusting settings. The user must enable Bluetooth permissions on their device and establish connection with agribot. The users can also memorize paths within the app, allowing for the efficient repetition of tasks without the need for constant manual input via remote. This innovation not only reduces workload but also enhances the overall efficiency of the system, aligning with the ethos of user-friendly precision. The application also facilitates automatic mode in which user can draw and give input directions to agribot for farming. During the execution phase, it is important to ensure that necessary operations such as seeding, sowing, and sprinkling are turned on/off as per need. This is to prevent any negative impacts on the farming process. Regular maintenance and updates are also essential to ensure that the system remains reliable and secure. In summary, the execution of the system and ShetiSakha require enabling Bluetooth permissions and establishing a connection with the system. Maintenance and updates are critical to ensure that the system remains reliable and secure.

7.3 Snapshots

7.3.1 Hardware Machine

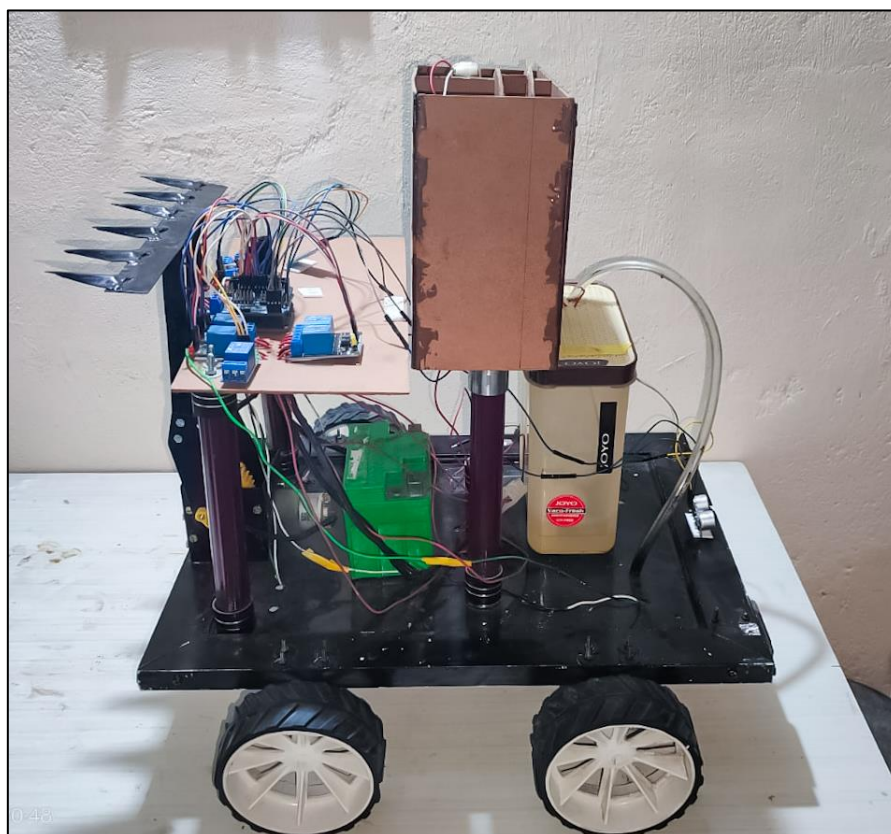
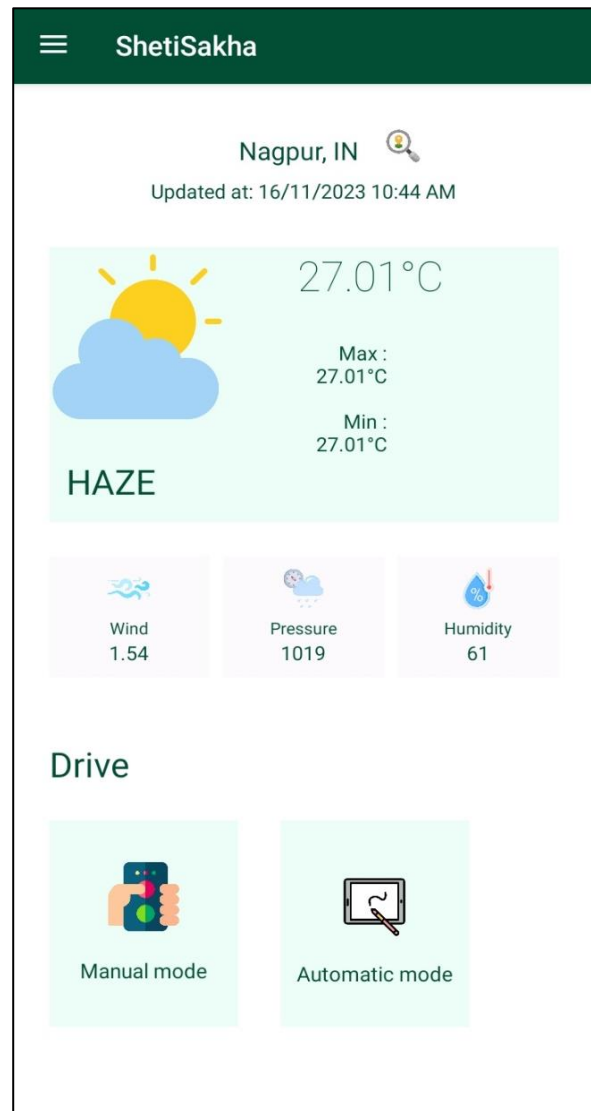
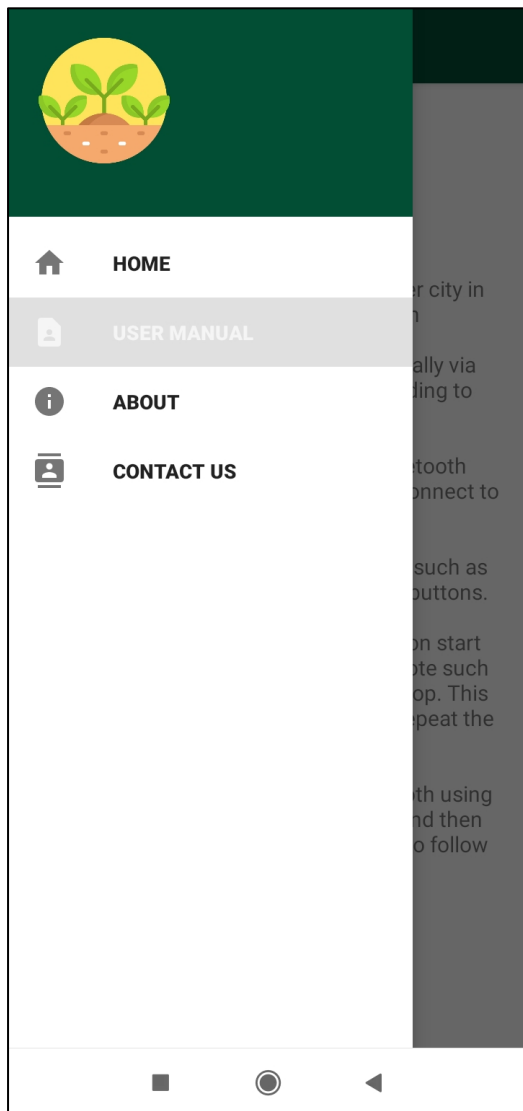
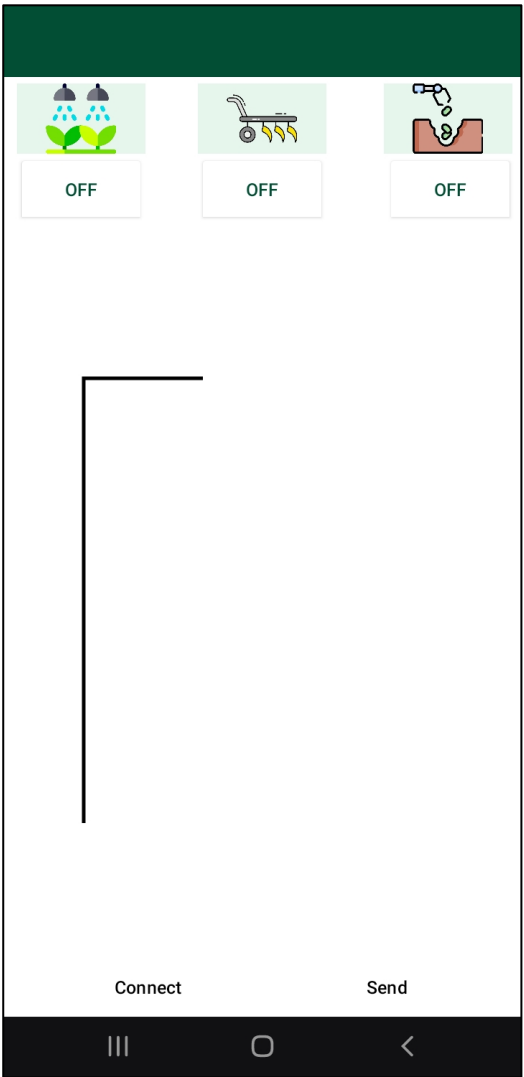
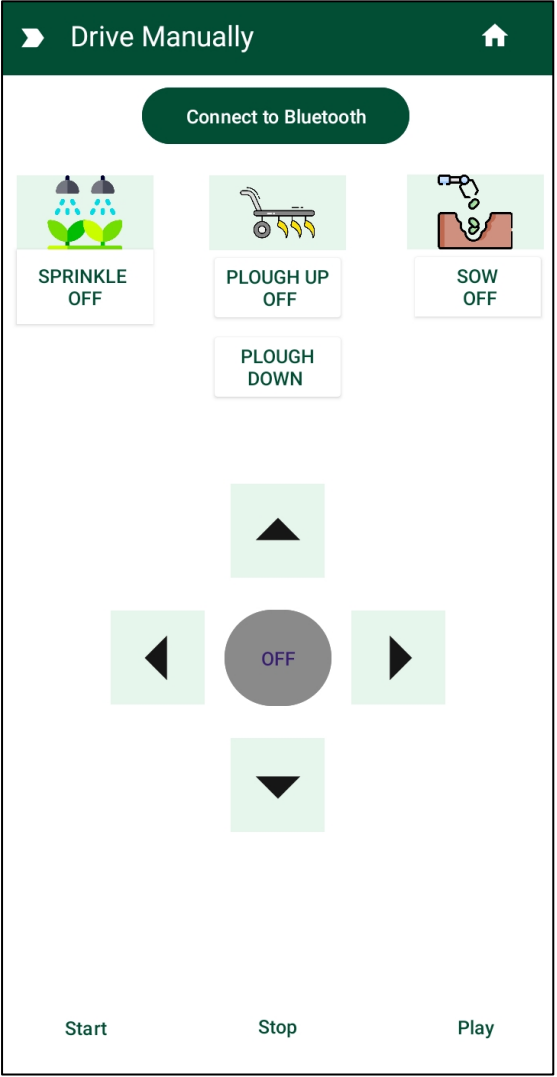
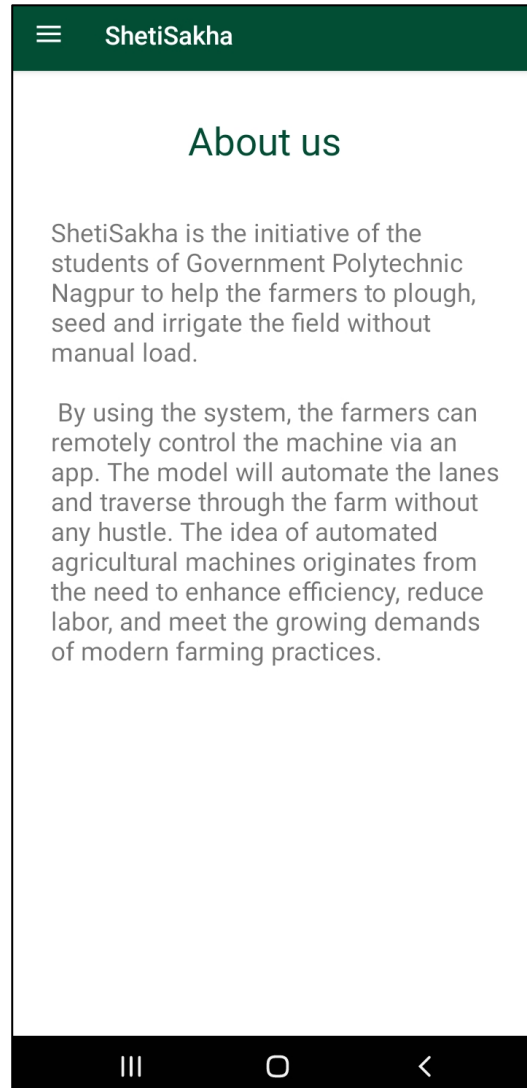
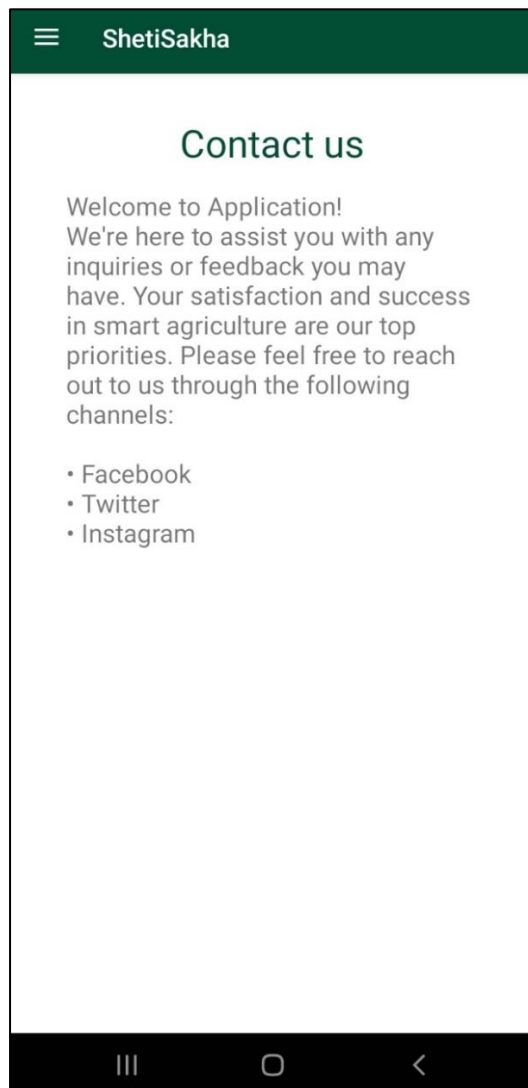


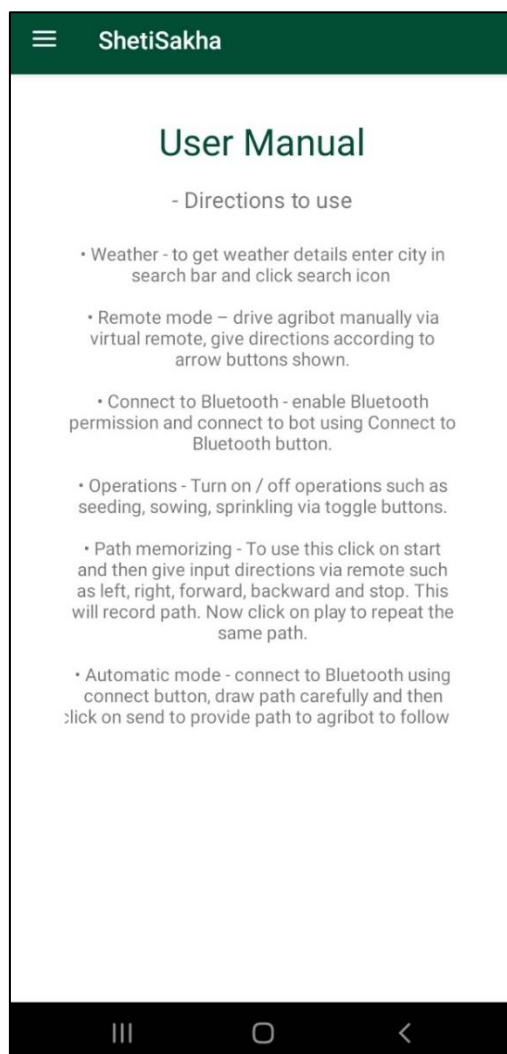
Fig 7.3.1 :Agribot Machine

7.3.2 Software Application Snapshot









Chapter 8

Conclusion

ShetiSakha is an innovative system designed to automate and optimize agricultural activities, ultimately improving the efficiency of production. In this study, we have developed a fully functional prototype of ShetiSakha and evaluated its performance in real farming conditions. The results of our testing have shown that ShetiSakha is an effective solution for automating the agricultural activities. The system significantly reduces the manual effort required for ploughing, sowing and sprinkling, and provides precise control over the traversal of the machine throughout the field. The integration of various hardware modules, such as the Bluetooth module, and Arduino UNO have enabled us to automate and make necessary adjustments in real-time. Through our testing, we compared the performance of ShetiSakha with the traditional manual agricultural operations. The results have demonstrated that ShetiSakha can produce crops at a faster rate, with minimal manual intervention. The automation for farm traversal has two modules- path memorizing where after once traversed through the field the path can be stored and in order to do the other operations the farmer does not have to constantly control the machine manually, instead by click of a button the same path can be traversed again, the only matter is it should have the same starting point as when path was saved. The other one is the draw-line mode where the functionalities of ploughing, sowing and sprinkling are to be switched on first and then on drawing the path on the app, the machine will move accordingly.

The automated system can significantly reduce the amount of energy required for these functionalities. In addition, the use of ShetiSakha can also help to reduce the amount of time taken to do these activities, by ensuring that the process is more efficient and effective. While ShetiSakha has shown great potential as a automated multipurpose agricultural machine, some limitations need to be addressed. One of the major limitations is the system may require some technical expertise to operate, which may be a challenge for novice users. In conclusion, ShetiSakha is a promising solution for automating agricultural activities, offering a range of benefits, including improved efficiency, quality, and environmental friendly. The system has the potential to revolutionize the way agricultural activities are carried out, by reducing manual effort, improving quality control, and promoting more sustainable practices. As such, we

believe that ShetiSakha has significant potential for commercialization and widespread adoption, with further refinement and development.

8.1 Suggestions

Based on the development, testing, and evaluation of ShetiSakha, the following recommendations can be integrated for improvement:

1. Precision Farming: sensors and cameras can be used to analyze the soil and weather patterns to provide a list of optimal decisions for the farmer to produce maximum output.
2. Computer Vision cameras: can help in detecting possible crop or soil diseases by identifying the symptoms and providing accurate solutions.
3. Increased Usability for the farmers: The application will have the option to select the native language to increase the usability of the application and make it convenient for the farmers.

Chapter 9

References

- 1) Multipurpose autonomous agricultural robot by IEEE
<https://ieeexplore.ieee.org/document/8212756>
- 2) Wikipedia: The free Encyclopaedia (the various Agriculture machine and related resources)
- 3) Farm bot Project Developed by Indian Institute of Science
http://www.kscst.iisc.ernet.in/spp/38_series/spp38s/synopsis_exhibition/154_38S1158.pdf
- 4) Blackmore, S. (2007). A systems view of agricultural robotics. Precision Agriculture conference, Wageningen Academic Publishers, the Netherlands. pp. 23- 31.
- 5) PATH MEMORIZING ALGORITHM
 - Real Time Navigation of Robot with Path Memorizing Algorithm
 - Urvi Patel¹, Narotam Sharma², Professor Vishal Vaidya³
 - 1,2,3 Dept. of Instrumentation & Control Engineering, Institute of Technology, Nirma University, Ahmadabad, Gujarat, India
 - Gregory Dudek, Michael Jenkin: Computational Principles of Mobile Robotics. Cambridge University Press 2000
- 6) DRAW LINE MODE
 - Driving Scenario Designer – By Matlab
<https://www.mathworks.com/help/driving/ref/drivingsceniodesigner-app.html>
 - Path Following Vehicle : <https://youtu.be/izkAtyGmQyA>

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